

Spend Smart: **AI-Powered** Finance Management

An intelligent platform for Budgeting, Analytics, and Automated Assistance

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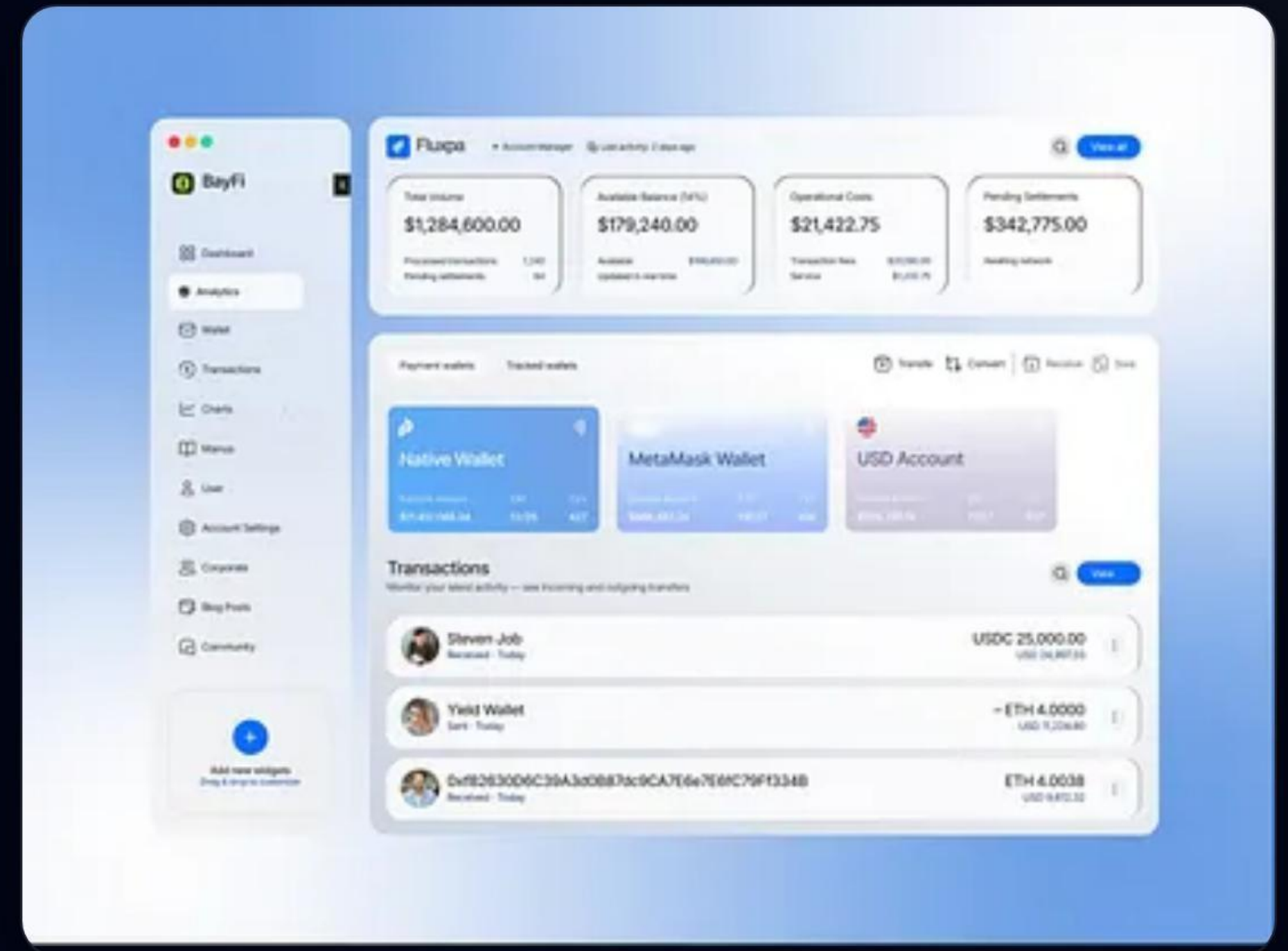
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| Executive Summary

The Vision

Modern finance tools must evolve from passive record-keeping to proactive assistance. Spend Smart is an all-in-one full-stack platform designed to bridge the gap between transactional data and actionable financial wisdom.

- ✓ Real-time income & expense tracking
- ✓ AI-driven behavioral analysis
- ✓ Automated reports & smart reminders



| Core System Objectives



Ease of Entry

Simplify transaction recording through categorization and OCR to reduce manual efforts.



AI Insights

Utilize Large Language Models to generate personalized financial advice and summaries.



Security

Maintain user trust via robust authentication and protection against malicious traffic.

| The Tech Stack: Frontend

Modern Web Excellence

Built with **Next.js 14** and **React**, the platform leverages Server-Side Rendering (SSR) for blazing performance.

Tailwind CSS: Rapid, utility-first styling.

Shadcn UI: Accessible, high-quality components.

Recharts: Interactive, vector-based data viz.

Lucide: Clean and consistent iconography.

 Full Stack Architecture Diagram

| Backend & Data Management

Prisma ORM

Type-safe database queries and automated schema migrations for PostgreSQL.

Supabase

Scalable cloud infrastructure hosting our PostgreSQL database and file buckets.

Inngest

Event-driven automation for recurring tasks, monthly summaries, and cron jobs.

Scalable architecture optimized for performance and data integrity.

The AI Engine: Gemini

Intelligent Automation

Spend Smart integrates **Google Gemini 1.5 Pro** to eliminate friction in financial management.

Receipt Scanning: Upload an image, and the AI extracts merchant, date, items, and taxes automatically into the database.

Narrative Reports: Monthly analysis transforms cold numbers into a human-readable financial story.



| Multi-Layered Security

 Cyber Security Lock Concept

Authentication

Clerk handles JWT sessions, OAuth, and Multi-Factor Authentication (MFA).

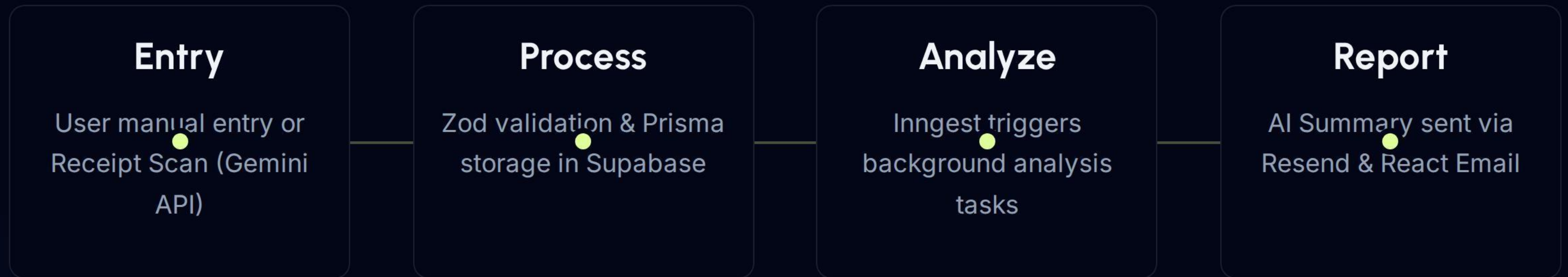
Abuse Mitigation

Arcjet middleware blocks bot traffic, prevents brute-force, and enforces rate limits.

Data Isolation

Supabase Row-Level Security (RLS) ensures users only see their own financial data.

| System Data Workflow



| Evaluation: Feature Comparisons

Traditional Apps

Basic OCR Only

FinSight AI

Cloud Based

Spend Smart

Gemini Multimodal + Automation

Our platform achieves maximum efficiency by leveraging Next.js Route-level caching and Prisma query optimization.

| System Comparison Matrix

Feature	Traditional App	FinSight AI	Spend Smart
AI Reports	None	Basic Charts	Gemini Generative
Automation	Manual	Inngest	Inngest (Cron + Events)
Bot Protection	None	Limited	Arcjet Middleware
File Storage	Local	Cloud	Supabase (Signed URLs)

| Limitations & Future Roadmap

Future Innovations

 **Open Banking:** Direct integration with bank feeds to eliminate all manual entry.

 **Conversational AI:** Natural language queries about financial status.

 **Predictive Analytics:** Forecasting future spending based on historical habits.

Current Milestone

Achieved full-stack integration with Gemini 1.5 Pro and Clerk security modules. Successfully automating 90% of the manual accounting workflow.

Thank You!

Questions & Feedback

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Presented by Team Spend Smart | Dept. of CSE (AI & ML) | KIET Group of Institutions

